

Ahmed Refat Abd El-Moaty

Data Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with hands-on experience designing and deploying scalable batch and real-time streaming pipelines, cloud-native data lakehouses, and production-grade analytics architectures. Proficient in Python, PySpark, SQL, Apache Kafka, Apache Airflow, Snowflake, dbt, and AWS/Azure cloud platforms. Skilled in Medallion Architecture (Bronze/Silver/Gold), Apache Iceberg, and dimensional data modeling — Maintaining end-to-end pipeline observability — consistently transforming high-volume raw data into reliable, analytics-ready assets that power business intelligence and machine learning at scale.

EDUCATION

B.Sc. Computer Science & Information Systems

Graduated: May 2023

Kafr El-Sheikh University, Faculty of Computers and Information · Information Systems Department

TECHNICAL SKILLS

Languages & Query	Python (OOP, Pandas, NumPy, Matplotlib, Seaborn), SQL, Bash / Linux
Big Data & Streaming	Apache Spark, PySpark, Apache Kafka, PySpark Structured Streaming, Hadoop / HDFS
Cloud Platforms	AWS (S3, Glue, Redshift, Athena, EC2) · Azure (ADLS, Data Factory, Synapse Analytics) · Snowflake, Databricks
Data Warehousing	Snowflake, Amazon Redshift, Azure Synapse Analytics, Apache Iceberg, TimescaleDB
Data Lakehouse Stack	Medallion Architecture (Bronze/Silver/Gold), Apache Iceberg, ACID Transactions, Schema Evolution
Orchestration & ETL	Apache Airflow, dbt, ETL/ELT Pipeline Design, Great Expectations (Data Quality)
Databases & Formats	PostgreSQL, MongoDB, SQL Server, Parquet, JSON
DevOps & CI/CD	Docker, Git / GitHub, Linux
Monitoring & BI	Power BI, Grafana
Architecture Patterns	Medallion Architecture, Lakehouse, Lambda Architecture, Star / Galaxy Schema, Dimensional Modeling

EXPERIENCE & TRAINING

Military Service (Mandatory National Service) — Egyptian Armed Forces

Dec 2023 – Mar 2025

Data Engineer Trainee — Microsoft – Digital Egypt Pioneer Initiative (DEPI)

Nov 2025 – May 2026

Cairo, Egypt · Microsoft-Certified Enterprise Data Engineering Program (6 Months)

- Engineered Python and SQL data pipelines across 2+ cloud-based projects on Azure, applying ETL/ELT patterns to ingest, transform, and deliver analytics-ready datasets — reducing data preparation time by 40% compared to manual processing baselines.
- Implemented Star and Galaxy Schema dimensional models across 3+ analytical domains, improving query structure consistency
- Completed a 6-month full-lifecycle program across 4 core engineering pillars — ingestion, transformation, warehousing, and orchestration — graduating with a Microsoft Data Engineer Certificate and delivering 2+ end-to-end cloud projects.

Cloud Services Trainee — National Telecommunication Institute (NTI)

Jan 2026 – Mar 2026

Egypt · Hands-On AWS Cloud Infrastructure Program (2 Months)

- Provisioned and configured 6+ AWS services — EC2, Identity & Access Management, Redshift, Glue, RDS, DynamoDB, and S3 — across 10+ lab environments, building operational readiness for cloud-native data engineering workflows.
- Applied Linux system administration and network fundamentals across 2 months of structured lab practice, reinforcing infrastructure skills critical for managing distributed data systems on AWS.

TECHNICAL PROJECTS

Financial Data Platform: AWS Cloud Data Warehouse [GitHub ↗](#)

AWS Glue (PySpark) · Amazon S3 · Amazon Redshift · Apache Airflow · dbt · Great Expectations · Power BI

- Architected a full Medallion Architecture (Bronze → Silver → Gold) ELT pipeline on AWS, processing 2M+ financial transaction records end-to-end — from raw S3 ingestion through fraud-detection-ready Gold layer datasets.
- Enforced data quality at the Silver layer using Great Expectations, automating schema validation, completeness checks, and accuracy assertions — preventing corrupt data from propagating to downstream analytics.
- Modeled a Star Schema data warehouse in Amazon Redshift using AWS Glue (PySpark) for transformation and dbt for business logic standardization, enabling structured fraud pattern analysis across transaction dimensions.
- Delivered a Power BI fraud detection dashboard connected to Redshift, with end-to-end pipeline orchestration via Apache Airflow DAGs ensuring reliable, scheduled data refresh.

NYC Smart Traffic: Batch & Streaming Analytics Platform [GitHub ↗](#)

Apache Kafka · PySpark Structured Streaming · Apache Airflow · Snowflake · TimescaleDB · Power BI · Grafana · AWS S3

- Engineered 2 fully independent data pipelines processing 40M+ records: a batch pipeline ingesting NYC traffic, weather, and incident data into a 3-layer S3 data lake (Raw → Processed → Curated) using PySpark, orchestrated by Apache Airflow.
- Modeled a Galaxy Schema data warehouse in Snowflake spanning 3 analytical domains (traffic, weather, incidents), delivering multi-dimensional historical analytics via Power BI dashboards.
- Built a parallel real-time streaming pipeline consuming live Citi Bike station data through Kafka → PySpark Structured Streaming → TimescaleDB, with a live Grafana dashboard tracking station availability and usage in real time.

Predictive Telecom Streaming Lakehouse [GitHub ↗](#)

Apache Iceberg · AWS S3 · Apache Kafka · PySpark Structured Streaming · Anomaly Detection (Isolation Forest) · Apache Airflow · Grafana · AWS Athena

- Architected a production-grade streaming Data Lakehouse on Apache Iceberg / AWS S3, ingesting 4 concurrent API streams (system, network, radio, environmental metrics) into dedicated Kafka topics for parallel real-time processing.
- Implemented a Bronze → Silver → Gold Medallion pipeline using PySpark Structured Streaming — Silver layer joins all 4 streams into unified Iceberg records on S3 with full ACID support, schema evolution, and automated compaction.
- Deployed an anomaly detection model (Isolation Forest) on each Gold layer micro-batch; decoupled a Python microservice consuming a dedicated Kafka topic to trigger Telegram Bot alerts — fully isolated from the core ML pipeline.
- Orchestrated 2 Airflow DAGs for periodic Iceberg compaction and anomaly model (Isolation Forest) retraining on historical lakehouse data, with live observability via Grafana connected to AWS Athena over the Gold layer.

Flight Delay Prediction Platform [GitHub ↗](#)

Apache Kafka · PySpark Structured Streaming · TimescaleDB · Grafana · XGBoost · Apache Airflow · Snowflake · Power BI

- Built a unified Streaming + Batch platform integrating 2 live APIs (OpenSky flight data, Open-Meteo weather) — streaming path routes data through Kafka → PySpark Structured Streaming → TimescaleDB → Grafana for real-time telemetry monitoring.
- Embedded an XGBoost Classifier directly into the streaming micro-batch pipeline for real-time flight delay prediction at 82% accuracy — delivering live ML inference alongside raw telemetry without a separate serving layer.
- Constructed a parallel batch pipeline with a 6-step Airflow DAG (Bronze → Silver → Gold → Snowflake), modeling a Galaxy Schema data warehouse and delivering 2 Power BI dashboards for historical delay pattern analysis.

CERTIFICATIONS

AWS Certified Solutions Architect · Amazon Web Services	Apr 2026
Microsoft Data Engineer Certificate · Digital Egypt Pioneer Initiative (DEPI)	May 2026
AWS Cloud Foundations · Amazon Web Services	Apr 2026
Cloud Services Management & Operation · National Telecommunication Institute (NTI)	Mar 2026
Linux Command Line & Docker · Udemy	Jan 2026
Analyzing & Visualizing Data with Power BI · Udemy	Dec 2025
Introduction to MongoDB · MongoDB University	Sep 2025

LANGUAGES

Arabic (Native) • **English** (Professional Working Proficiency — B2+)